MATLAB | karthikkumarp2416sse-bit/5G-... | MIMO System BER Analysis | (28) WhatsApp | Ask Gemini

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Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid | LABELS AND ANNOTATIONS

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Files

Name

- 5G AND 6G LAB
- LAB EXPERIMENT
- New Folder
- ASSEMENT QNO 1.m
- EX : 2.m
- ex 1.m
- EX 2 ..m
- EX 2..m
- Ex 2.m
- ex 5.m
- ex1.m

Workspace

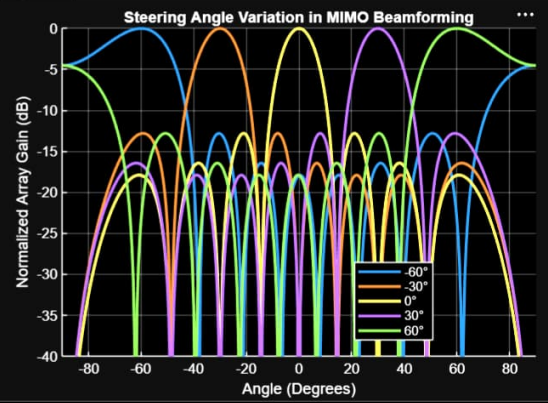
Name	Value	Size	Class
AF	1×361 double	1×361	double
k	361	1×1	double
m	5	1×1	double
N	8	1×1	double
phase	[0;0.4209;0.8...	8×1	double
steeringA...	[-60,-30,0,30,...	1×5	double
theta	1×361 double	1×361	double
theta0	60	1×1	double

2.m | ex3.m | EX4.m | untitled4.m | EX6.m | EX7.m | EX8.m

/MATLAB Drive/EX8.m

```
17        for k = 1:length(theta)
18
19            phase = pi*(0:N-1)'*(sind(theta(k)) - sind(theta0));
20            AF(k) = abs(sum(exp(1j*phase)));
21
22        end
23
24        % Normalize
25        AF = AF/max(AF);
26
27        % Plot in dB
28        plot(theta,20*log10(AF),'LineWidth',2);
29
30    end
31
32    grid on;
33    xlabel('Angle (Degrees)');
34    ylabel('Normalized Array Gain (dB)');
35    title('Steering Angle Variation in MIMO Beamforming');
36    legend('-60°','-30°','0°','30°','60°','Location','best');
37    axis([-90 90 -40 0]);
```

Figure 1



Command Window

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